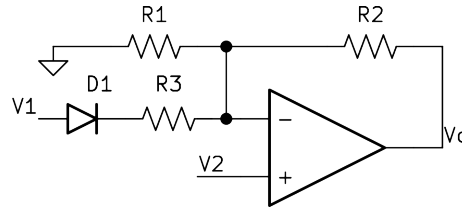
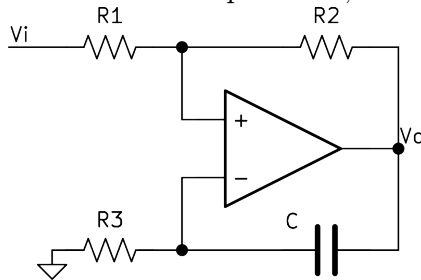


Q.1 [25]: Given the non-linear amplifier circuit below, what is the full relationship for V_o in terms of V_1 and V_2 for any combination of input voltages? Be sure to specify the limits to any voltage ranges.



Q.2 [25]: Using the above schematic, derive expressions for the input and output *noise spectral densities*, $e_{ni}(jf)$ and $e_{no}(jf)$, and the total rms output noise, E_{no} , between 100 mHz and 100 kHz. Assume $V_1 = 3\text{ V}$ and $V_2 = -3\text{ V}$ (both DC), the system is at 25°C , $R_1 = 10\text{ k}\Omega$, $R_2 = 20\text{ k}\Omega$, $R_3 = 5\text{ k}\Omega$, the diode has a forward voltage of 0.7 V , and the diode noise is modeled as simple shot noise: $i_n^2 = 2qI_D$, where q is the electron charge, $1.603 \times 10^{-19}\text{ C}$. The op-amp noise is characterized to have the values $e_{nw} = 30\text{ nV}/\sqrt{\text{Hz}}$, $f_{ce} = 50\text{ Hz}$, $i_{nw} = 100\text{ pA}/\sqrt{\text{Hz}}$, and $f_{ci} = 0\text{ Hz}$. You may assume that $f_B \gg 100\text{ kHz}$.

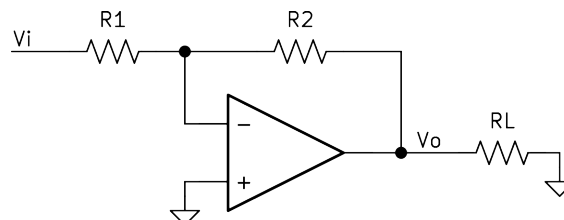
Q.3 [35]: For this set of questions, consider the circuit below with both positive and negative feedback.



$$T(jf) = a(jf) \frac{R_1}{R_1 + R_2} \frac{1 - j2\pi f \frac{R_2 R_3}{R_1} C}{1 + j2\pi f (R_3 + r_o) C}$$

- Show that the return ratio for the circuit results in the symbolic expression provided on the right. Assume that $r_o \ll R_1, R_2$. [20]
- Note that the return ratio has a *negative zero*, something we did not handle in this class; however, the effect of this behavior may be deduced by taking frequency limits, $f \rightarrow 0$ and $f \rightarrow \infty$. What are these limits for β (simplify) and how does this result differ from if the zero was positive, specifically with respect to the magnitude and phase of β . [10]
- Assuming the amplifier has a single dominant pole at f_b , how would you determine the range of C values for which system is stable? Note: since we have a negative zero, rate-of-closure analysis is not valid, at least how it was learned in class. Set up the equations, etc., that you would use but do not solve. If approaching via a modified rate-of-closure analysis, draw the appropriate figure(s) and note important differences. [5]

Q.4 [15]: For the circuit below, find the impedance experienced by resistor R_2 in the circuit below. $r_d \neq \infty$ and $r_o \neq 0$. The amplifier has a constant open loop gain of $a(jf) = a_o$. Draw all relevant figures.



This exam is for both:

ECSE-4050 and ECSE-6050

Flip this page over only when instructed to do so.

This exam is open book, open notes. You may use your computer/tablet only to view course content. This includes and is limited to: the textbook, lecture notes (both personal and provided), homeworks, and homework solutions. All other uses are prohibited. Typing on the computer/tablet is not allowed; excluding for scrolling through documents.

For each problem: mark the problem number and all subquestions clearly and box your final answer(s).

Answers that are not supported by work will not receive credit.

All answers will be written on blank paper, provided as necessary.

Write only on one side.